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Oops Test

1. Create a vehicle class without any Variables and Methods.

```
Class vehicle :
      pass.
```

2. Create a child class Bus that inherill from vehicle.

```
Class vehicle :
      Pass

clear Bus(vehicle):
      Pass

b = Bus()
print(" Bus class created successfully")
```

3. Define a property that must have the same value for every class instance (object) Define a class attribute "color" with a default value white.

Code :

```
class Vehicle :
    color = "white"

class Bus (vehicle):
    pass

class Car (vehicle):
    pass

bus = Bus()
car = Car()
print ("Bus color:", bus.color)
print ("car colour:", car.color)
```

A) class attribute country with default value "USA";

```
class Person :
    country = "USA"

class Student (Person):
    pass

class Teacher ( Person):
    pass

s = Student ()
t = Techer ()
print ("student country:", s.country)
print (" Teacher country:", t country).
```

5. Bank class with class attribute currency.

```
class Bunk :
    currency ="USD"
class SavingsAccount (Bank):
    Pass.
class checking Account (Bank):
    currency ="INR"

s = Saving Account()
c = Checking Account()
Print ("Savings Account currency:", s.currency)
Print ("Checking Account currency:", c.currency)
```

6. Car class with private attribute __engine_number.

```
class car :
    def __init__(self, engine_number):
        self.__engine_number = engine_number.

    def get_engine_number( self):
        return self.__engine_number

car = Car ("ENG12345")
Print ("Engine Number :", car.get_engine_number())
```

7) Abstract class Animal with speak()

```
from abc import ABC, abstract method

class Animal (ABC):
    @abstractmethod
    def speak (self):
        pass

class Dog (Animal):
    def speak (self):
        print ("Dog says: Bark")

class cat (Aimal):
    def speak (self):
        print("Cat says : Meow")

d = Dog()
c = Cat()

d. speak()

c. speak()
```

8) vehicle class with instance attributes:

```
class vehicle:
    def __init_(self, max_speed, mileage):
        self.max_speed = max-speed
        self. mileage = mileage

car. = vehicle (180, 20)
Print ("Max", car.max. speed)
Print ("Milege", car. Mileage)
```

9) Take a three digit number and add the Square of each digit.

```
num = input (" Enter 3 digit number:")
total = 0
for digit in num:
        total += int (digit) ** 2.
print (" Sum of square", total).
```